Rituraj Singh

Curriculum Vitae

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Email	rituraj singh.infotech@gmail.com
Google Scholar	Rituraj Singh
Website	https://devrituraj.github.io/
Languages	Maithili (native), Hindi (fluent), English (fluent), Bengali (beginner), French (beginner).

RESEARCH AREAS

My PhD research focuses on developing algorithms for crowdsourcing applications: to propose formal model for complex workflows for crowdsourcing and developing human-in-loop algorithms by applying probabilistic based techniques and latent variable models. I am also passionate about applying machine learning methods in other domains as healthcare, natural language processing, etc. Previously, I did some work in anomaly detection on sensor data and on detection of diseases based on data obtained using wearable. My area of interest are Data Centric Systems and Workflows, Machine Learning, Crowdsourcing, Probabilistic Modelling of Systems and Data Mining.

EDUCATION

PhD in Computer Science & Engineering 2018-Now
• *INRIA* • *IRISA* • *University de Rennes 1, France*

- Co-advised by Dr. Loïc Hélouët and Dr. Zoltán Miklós
- Thesis: Data Centric Workflows for Crowdsourcing Applications.
- Thesis submitted to rapporteurs and defense expected in April 2021.

M.Tech in Computer Science & Engineering 2013-2015
• *Indian Institute of Technology, Patna, India.*

- Advised by Dr. Ashok Singh Sairam
- CGPA: 8.79 (10.0 scale)
- Thesis: Push based User Selection in Crowdsensing.

B.Tech. in Information Technology 2008-2012
Rajasthan Technical University, Rajasthan, India.

- Percentage: 74.28
- Pass with Honours

WORK EXPERIENCE

Researcher 2015-2017
TCS Research and Innovation Labs, Kolkata, India

- Research on applying machine learning models on sensor and wearable healthcare data. The work led to several publications and patents.

Internship Summer 2014
TCS Research and Innovation Labs, Kolkata, India

- Project on anomaly detection using unsupervised learning in Smart meter sensor data.

RESEARCH INTERESTS

Machine learning, Crowdsourcing, Data Centric Systems and Workflows, Signal processing, Pattern recognition, Probabilistic Modelling of Systems, Deep learning, Databases, Data Mining, Artificial intelligence.

PATENTS

10. Soma Bandyopadhyay, Arijit Ukil, Rituraj Singh, Puri Chetanya, Arpan Pal, CA Murthy , "Anomaly detection by self-learning of sensor signals", Patent No. : US 10,743,821 B2

9. Sahu Ishan, Ayan Mukherjee, Arijit Ukil, Soma Bandyopadhyay, Puri Chetanya, Rituraj Singh, Arpan Pal, Rohan Banerjee , "Method and system for joint selection of a feature subset-classifier pair for a classification task", Pub. No. : US 2019/0361919 A1

8. Soma Bandyopadhyay, Arijit Ukil, Puri Chetanya, Rituraj Singh, Arpan Pal, C A Murthy, "Method and system for pattern recognition in a signal using morphology aware symbolic representation", Pub. No. : US 2019/0278971 A1

7. Soma Bandyopadhyay, Arijit Ukil, Puri Chetanya, Rituraj Singh, Arpan Pal, C A Murthy, "Systems and methods for detecting anomaly in a cardiovascular signal using hierarchical extremas and repetitions", Pub. No. : US 2019/0200935 A1

6. Shreyasi Datta, Puri Chetanya, Ayan Mukherjee, Rohan Banerjee, Anirban Dutta Choudhury, Arijit Ukil, Soma Bandyopadhyay, Arpan Pal, Sundeep Khandelwal, Rituraj Singh, "Cascaded binary classifier for identifying rhythms in a single-lead electrocardiogram (ECG) signal", Patent. No. : US US 10,750, 968 BB2

5. Arijit Ukil, Soma Bandyopadhyay, Puri Chetanya, Rituraj Singh, Arpan Pal, "Generalized one-class support vector machines with jointly optimized hyperparameters", Pub. No. : US 2019/0050690 A1

4. Arijit Ukil, Soma Bandyopadhyay, Puri Chetanya, Rituraj Singh, Arpan Pal, "Synthetic rare class generation by preserving morphological identity", Pub. No. : US 2019/0050673 A1

3. Soma Bandyopadhyay, Arijit Ukil, Puri Chetanya, Rituraj Singh, Arpan Pal, CA Murthy, Kayapanda Mandana, "Method and system for physiological parameter derivation from pulsating signals with reduced error", Patent No.: US 10,172,528 B2

2. Arijit Ukil, Soma Bandyopadhyay, Puri Chetanya, Rituraj Singh, Arpan Pal, Debayan Mukherjee, "System and method for physiological monitoring", Pub. No.: US 2018/0153419 A1

1. Arijit Ukil, Soma Bandyopadhyay, PURI Chetanya, Arpan Pal, Rituraj Singh, Ayan Mukherjee, Debayan Mukherjee, "Noisy signal identification from non-stationary audio signals", Patent. No. : US 9,978,392 B2

RESEARCH PUBLICATIONS

8. Loïc Hélouët, Zoltan Miklos, Rituraj Singh, "Cost and Quality Assurance in Crowdsourcing Workflows", Submitted.

7. Rituraj Singh, Loïc Hélouët, Zoltán Miklós, "Reducing the Cost of Aggregation in Crowdsourcing", International Conference on Web Services (ICWS 2020).

6. Pierre Bourhis, Loïc Hélouët, Rituraj Singh, Zoltán Miklós, "Data Centric Workflows for Crowdsourcing", 41st International Conference on Application and Theory of Petri Nets and Concurrency, (PetriNets) 2020.

5. Arijit Ukil, Ishan Sahu, Chetanya Puri, Ayan Mukherjee, Rituraj Singh, Soma Bandyopadhyay, Arpan Pal, "AutoModeling: Integrated Approach for Automated Model Generation by Ensemble Selection of Feature Subset and Classifier", International Joint Conference on Neural Networks (IJCNN) 2019

4. Arijit Ukil, Soma Bandyopadhyay, Chetanya Puri, Rituraj Singh, Arpan Pal, "Effective Noise Removal and Unified Model of Hybrid Feature Space Optimization for Automated Cardiac Anomaly Detection Using Phonocardiogarm Signals", IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2018.

3. Soma Bandyopadhyay, Arijit Ukil, Rituraj Singh, Chetanya Puri, Arpan Pal, CA Murthy, "Pattern Analysis in Physiological Pulsatile Signals: An Aid to Personalized Healthcare" 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) 2018

2. Arijit Ukil, Soma Bandyopadhyay, Chetanya Puri, Rituraj Singh, Arpan Pal, "On Solving the Class Imbalance Problem for Clinical Decision Improvement Using Heart Sound Signals",Proceedings of the IJCAI 2017 - Workshop on Learning in the Presence of Class Imbalance and Concept Drift

1. Soma Bandyopadhyay, Arijit Ukil, Rituraj Singh, Chetanya Puri, Arpan Pal, CA Murthy, "3S: Sensing Sensor Signal ",Proceedings of the 14th ACM Conference on Embedded Network Sensor Systems (SenSys) 2015

   Note: For complete list, please see here.

ACHIEVEMENTS

- Fellowship - ANR Headwork Project (2018).
- Scholarship - GATE Masters Scholarship (2013-2015).
- Received "TCS i-Citation Award" (2016 & 2017).
- Winner of Physionet Challenge (2017).
- All India Rank 440 (out of 224160 candidates) in Graduate Aptitude Test in Engineering (GATE) 2013.

TEACHING EXPERIENCE

Travaux Pratiques[1] for Artificial Intelligence - Topics on Deep Learning Spring 2021

- *Graduate level introduction to Deep learning (Fully Connected Network, CNN, RNN, LSTM) using Tensorflow, Keras at ESIR — University of Rennes 1.*
- *Course taught by Dr. Zoltán Miklós.*

Travaux Pratiques Data Mining Course Spring 2019, Spring 2018

- *Graduate level course on data mining using numpy, pandas, scipy, sklearn libraries at ESIR — University of Rennes 1.*
- *Course taught by Dr. Zoltán Miklós.*

Teaching Assistant for JAVA programming language Fall 2014

- *Course taught to under graduate third year students at IIT Patna. The objective of the course was to learn JAVA programming.*
- *Course taught by Dr. Ashok Singh Sairam.*

Teaching Assistant for Shell programming and C programming language Spring 2013

- *Course taught to under graduate first year students at IIT Patna. The objective of the course was to have the basic understanding of shell programming and C language.*
- *Course taught by Dr. Raju Halder.*

[1] Equivalent to Teaching Assistant

COMPUTER SKILLS

- Programming languages: Python, Java, C, C++, Android.
- Frameworks: Tensorflow, NumPy, SciPy, Sklearn, Pandas.
- Database Systems: MySQL, SPARQL.
- Scientific: MATLAB, LATEX, Git.

PROJECTS

- Data Centric Workflows for Complex Crowdsourcing Applications.
- Crowdsensing using Mobile Phone.
- Anomaly detection and its impact on disease classification on Bio-medical data.
- Machine Learning projects on image classification, sentiment analysis and Natural language processing.

Note: For further details on my research, please visit the publications page.

EXTRA CURRICULUM ACTIVITIES

- Sports: Badminton, video-games, hiking.
- Hobbies: traveling, reading, movies, playing keyboard.

REFERENCES

1. Dr. Loïc Hélouët, Chargé de recherche, SUMO Team, INRIA, Rennes, France.
2. Dr. Zoltán Miklós, Assistant Professor, Universite de Rennes 1, DRUID Team, IRISA, Rennes, France.
3. Dr. Ashok Singh Sairam, Associate Professor,Indian Institute of Technology, Guwahati, India.